



Semester One Examination, 2024

Question/Answer booklet

**MATHEMATICS
METHODS
UNIT 1**

**Section One:
Calculator-free**

If required by your examination administrator, please
place your student identification label in this box

SOLUTIONS

Time allowed for this section

Reading time before commencing work: five minutes
Working time: fifty minutes

Number of additional
answer booklets used
(if applicable):

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet
Formula sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener,
correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	6	6	50	52	35
Section Two: Calculator-assumed	12	12	100	98	65
Total					100

Instructions to candidates

- The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
- Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
- You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
- Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
- It is recommended that you do not use pencil, except in diagrams.
- Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
- The Formula sheet is not to be handed in with your Question/Answer booklet.

Markers use only		
Question	Maximum	Mark
1	6	
2	9	
3	12	
4	11	
5	7	
6	7	
Section 1 Total	52	
Weighted ($\times 0.6731$)	35%	
Section 2 Weighted	65%	
Total	100%	

Section One: Calculator-free

35% (52 Marks)

This section has **six** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

Question 1

(6 marks)

(a) Evaluate ${}^{100}C_{98}$

(2 mark)

Solution
${}^{100}C_{98} = \frac{100!}{98! \times 2!}$ $= \frac{100 \times 99}{2} \quad \checkmark$ $= 4950 \quad \checkmark$
Specific behaviours
✓ convert factorials ✓ correct answer

(b) Determine the value of m ($m \geq 0$) in each of the following equations.

(3 marks)

(i) $\binom{m}{0} + \binom{m}{1} + \binom{m}{2} + \binom{m}{3} = 2^m$

(1 mark)

Solution
$m = 3 \quad \checkmark$

(ii) $\binom{15}{2m} - \binom{15}{m+3} = 0$

(3 marks)

Solution
$\binom{15}{2m} = \binom{15}{m+3} \quad \checkmark$ $\Rightarrow 2m + m + 3 = 15 \quad \checkmark \quad \text{or} \quad 2m = m + 3 \quad \checkmark$ $\Rightarrow m = 4 \quad \text{or} \quad m = 3 \quad \checkmark$
Specific behaviours
✓ moves to equate ✓ states one equations ✓ States two solutions

Question 2

(9 marks)

(a) Solve the following equations.

(i) $x^2 = 49$.

(1 mark)

Solution
$x^2 - 49 = 0 \rightarrow x = -7, x = 7$ ✓
Specific behaviours
✓ correct solutions (both) accept ± 7

(ii) $2x^2 = 5x$

(1 mark)

Solution
$x = 0, x = 2.5$ ✓
Specific behaviours
✓ correct solutions (both)

(iii) $3x^2 + 11x - 4 = 0$.

(2 marks)

Solution
$(3x - 1)(x + 4) = 0$ ✓ $\rightarrow x = \frac{1}{3}, x = -4$ ✓
Specific behaviours
✓ factorises ✓ correct roots

- (b) Determine the coordinates of the turning point of the graph of $y = (x + 2)(x - 8)$. (2 marks)

Solution
Midpoint of roots: $x = (-2 + 8) \div 2 = 3$. ✓
When $x = 3$, $y = (5)(-5) = -25$.
Turning point is at $(3, -25)$. ✓
Specific behaviours
✓ correct x -coordinate ✓ correct ordered pair

- (c) The equation of the circle with centre $(-2, 0)$ and radius 3 is $x^2 + y^2 + ax + b = 0$. Determine the value of the constant a and the value of the constant b . (3 marks)

Solution
Factored equation of circle is: $(x + 2)^2 + (y - 0)^2 = 3^2$. ✓
Expanding we obtain: $x^2 + y^2 + 4x + 4 - 9 = 0$. ✓
Hence $a = 4$ and $b = -5$. ✓
Specific behaviours
✓ substitutes and expresses as $(x + 2)^2 + (y - 0)^2 = 3^2$ ✓ expands ✓ states both correct values

Question 3

(12 marks)

100 high school students were asked what pets they have. The results are as follows.

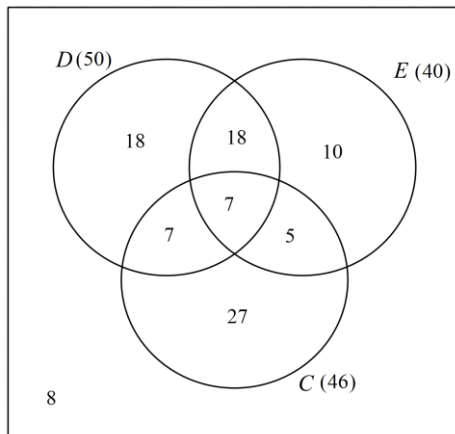
50 have a dog (D)
25 have a dog and birds

40 have birds (B)
14 have a dog and cat
7 have all three

46 have a cat (C)
12 have a cat and birds

(a) Represent this information in the Venn diagram below.

(3 marks)



Specific behaviours	
✓	Places 3 correct values
✓	Places 6
✓	Places 8

(b) Describe the students in the set $(D \cup B) \cap C'$.

(1 mark)

Solution
Students who have either a dog or birds (or both) but not a cat ✓
Specific behaviours
✓ correct solution

(c) Calculate the probability that a student has exactly two types of pets.

(1 mark)

Solution
$\frac{18 + 7 + 5}{100} = 0.3$ ✓
Specific behaviours
✓ correct solution

(d) Calculate the probability that a student has no pets.

(1 mark)

Solution
$\frac{8}{100} = 0.08$ ✓
Specific behaviours
✓ correct solution

- (e) Calculate the probability that a student has a dog given that they have only one type of pet. (2 marks)

Solution
$\frac{18}{18 + 10 + 27} = \frac{18}{55}$ ✓✓
Specific behaviours
✓ correct numerator or denominator
✓ correct solution

One of the students claimed that owning a cat has no impact on whether you own a bird.

- (f) Comment on this claim, including suitable calculations to justify your response. (4 marks)

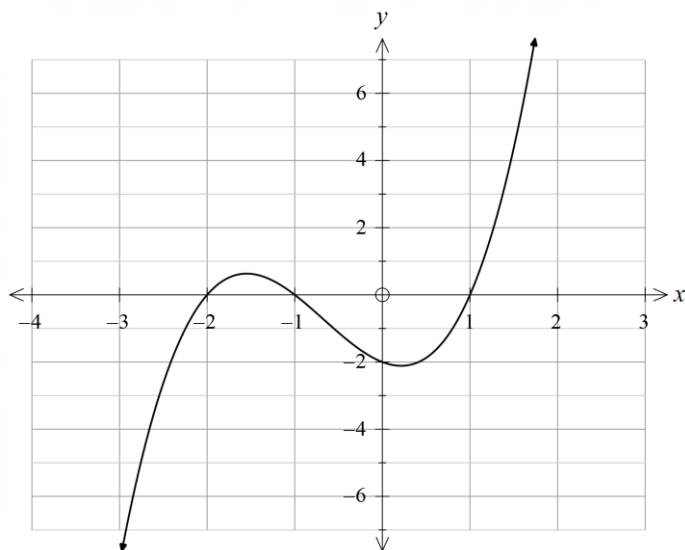
Solution
$P(cat) = \frac{46}{100} = \frac{23}{50}$ $P(bird) = \frac{40}{100} = \frac{2}{5}$ ✓ both
$P(cat \text{ and } bird) = \frac{12}{100} = \frac{3}{25}$ ✓ $P(cat) \times P(bird) = \frac{23}{50} \times \frac{2}{5} = \frac{23}{125} \neq \frac{3}{25}$ ✓ or $P(c / b) = \frac{12}{40} \neq P(c)$ ✓✓ or $P(b / c) = \frac{12}{46} \neq P(b)$
As $P(cat) \cdot P(bird) \neq P(cat \text{ and } bird)$ the events of having a cat and having birds is not independent Therefore the claim is false ✓
Specific behaviours
✓ Determines P(Cat) and P(Birds)
✓ Determines P(Cat and Birds) or ✓✓ $P(c / b) = \frac{12}{40} \neq P(c)$
✓ Demonstrates that $P(c).P(b)$ is not equal to $P(c \text{ and } b)$
✓ Makes statement that events are NOT independent , there claim is false

Question 4

(11 marks)

(a) Determine, in factorised form, the function $f(x)$ shown on the graph below.

(2 marks)

**Solution**

From the graph:

$$f(x) = a(x + 1)(x - 1)(x + 2) \checkmark$$

Substituting $(0, -2)$

$$-2 = -2a$$

$$a = 1$$

$$\therefore f(x) = (x + 1)(x - 1)(x + 2) \checkmark$$

Specific behaviours

- ✓ Writes $f(x)$ as a product of three factors.
- ✓ **Determines vertical dilation** and writes down $f(x)$

(b) Expand the function $f(x)$ found in part (a).

(2 marks)

Solution

$$f(x) = (x^2 - 1)(x + 2) \checkmark$$

$$f(x) = x^3 + 2x^2 - x - 2 \checkmark$$

Specific behaviours

- ✓ Expands two linear factors correctly.
- ✓ Correctly expands $f(x)$.

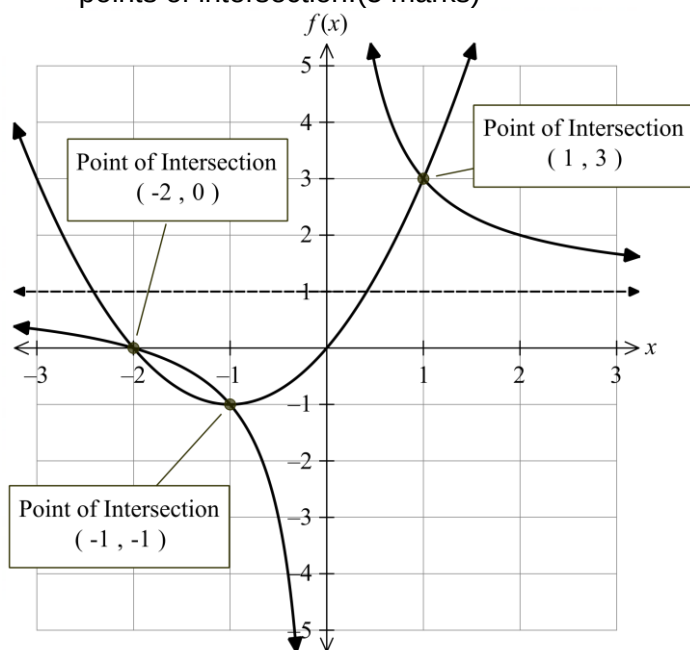
A student is solving the following simultaneous equations:

$$y = x^2 + 2x \quad y = \frac{2}{x} + 1$$

- (c) Show that the student would need to solve $x^3 + 2x^2 - x - 2 = 0$. (2 marks)

Solution
$x^2 + 2x = \frac{2}{x} + 1$ $x^2 + 2x - 1 = \frac{2}{x}$ $x^3 + 2x^2 - x = 2$ $x^3 + 2x^2 - x - 2 = 0$
Specific behaviours
✓ Equates two functions ✓ Correctly rearranges equation

- (d) Sketch on the axes below the graphs of $y = x^2 + 2x$ and $y = \frac{2}{x} + 1$, clearly labelling all points of intersection. (5 marks)

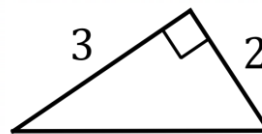


Specific behaviours
For graph of $y = x^2 + 2x$: ✓ Correct x-intercepts. ✓ Correct turning point and graph. For graph of $y = \frac{2}{x} + 1$: ✓ Horizontal asymptote drawn. ✓ Graph is asymptotic to $y = 1$ and y-axis. ✓ Both graphs intersect at the given points of intersection and labelled

Question 5

(7 marks)

- (a) Using the right triangle shown, or otherwise, if $\tan \lambda = 1.5$ and λ is an acute angle, determine the exact value of $\sin \lambda$.



(2 marks)

Solution
Using ratio of sides, $\tan \lambda = 1.5 = \frac{3}{2}$, and length of hypotenuse is $\sqrt{3^2 + 2^2} = \sqrt{13}$. ✓
Hence $\sin \lambda = \frac{3}{\sqrt{13}}$. ✓
Specific behaviours
✓ numerator
✓ denominator

- (b) Determine all solutions of the equation $4 \cos(x - 11^\circ) - 2\sqrt{2} = 0$ for $0^\circ \leq x \leq 360^\circ$. (3 marks)

Solution
$4 \cos(x - 11^\circ) = 2\sqrt{2}$ $\cos(x - 11^\circ) = \frac{\sqrt{2}}{2}$ ✓ $x - 11^\circ = 45^\circ, 315^\circ$ $x = 56^\circ, 326^\circ$ ✓
Specific behaviours
✓ obtains expression for $\cos(x - 11^\circ)$ ✓ determines one angle for $x - 11^\circ$ ✓ obtains both correct solutions

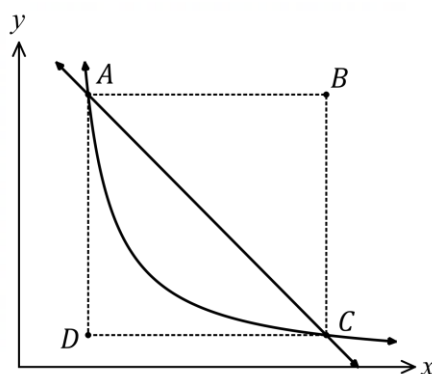
- (c) Determine the equation of the straight line that passes through the point $(0, -2)$ and has an angle of inclination of 30° with the positive x -axis. (2 marks)

Solution
Gradient of line is $\tan(30^\circ) = \frac{1}{\sqrt{3}}$. ✓ Equation of line is $y = \frac{1}{\sqrt{3}}x - 2$. ✓
Specific behaviours
✓ correctly relates angle of inclination to gradient ✓ correct equation

Question 6

(7 marks)

The diagram, not drawn to scale, shows rectangle $ABCD$ with side AB parallel to the x -axis, so that A and C are the points of intersection of the line $y = 9 - x$ with the hyperbola $y = \frac{6}{x-1}$.



- (a) State the equations of the two asymptotes of the hyperbola. (2 marks)

Solution
Asymptotes are $x = 1$ ✓ and $y = 0$. ✓
Specific behaviours
✓ vertical asymptote
✓ horizontal asymptote

- (b) Determine the area of rectangle $ABCD$. (5 marks)

Solution
Points of intersection: $\frac{6}{x-1} = 9 - x$ $6 = (x-1)(9-x)$ $6 = 9x - x^2 - 9 + x$ $x^2 - 10x + 15 = 0$ ✓ $(x-5)^2 - 25 + 15 = 0$ ✓ $(x-5)^2 = 10$ $x-5 = \pm\sqrt{10}$ $x = 5 \pm \sqrt{10}$ ✓ $y = 9 - (5 \pm \sqrt{10}) = 4 \mp \sqrt{10}$ ✓
Width of rectangle: $w = (5 + \sqrt{10}) - (5 - \sqrt{10}) = 2\sqrt{10}$ Height of rectangle: $h = (4 + \sqrt{10}) - (4 - \sqrt{10}) = 2\sqrt{10}$ Area of rectangle: $A = 2\sqrt{10} \times 2\sqrt{10} = 40$ sq units ✓
Specific behaviours
✓ forms equation and simplifies to obtain correct quadratic ✓ indicates use of quadratic formula or completion of square ✓ correctly solves quadratic for x ✓ obtains y -coordinates ✓ correct area of rectangle

End of questions

See next page

Supplementary page

Question number: _____